

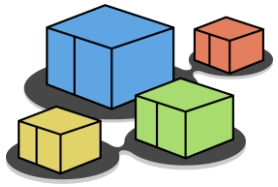
Kathará

Kathará lab

basic IPv6 configuration, ping, traceroute and ICMPv6

Version	3.0
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Web	http://www.kathara.org/
Description	basic IPv6 configuration commands, IPv6 stateless auto-configuration, usage of ping and traceroute, ICMPv6 behaviour

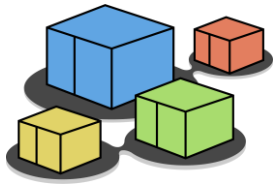
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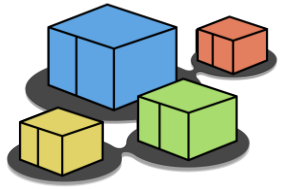
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content of the lab

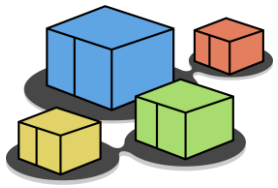
- there are two routers, called r1 and r2, and three hosts, called pc1, pc2, and pc3
  - they are connected via three LANs
  - we force their MAC addresses to be easily readable
- we will learn how to:
  - administratively assign an IPv6 address and a netmask to the interface of a system
  - administratively assign a default gateway to the interface of a system
  - set the IPv6 addresses of a group of end-systems using the MAC-address-based stateless-auto configuration
  - set the routing table of a router
- we will use the ping and traceroute commands
- we will observe the behavior of ICMPv6



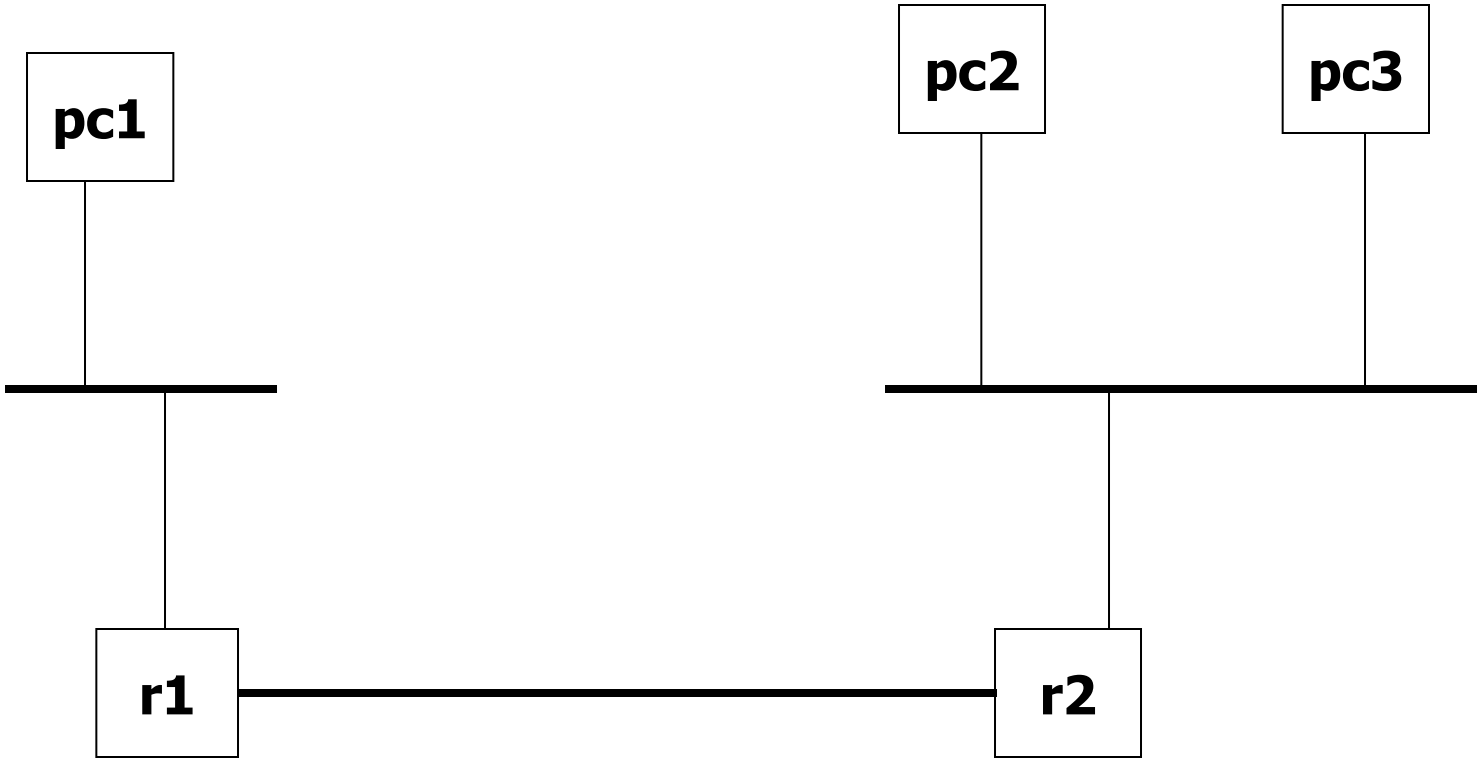
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lab configuration

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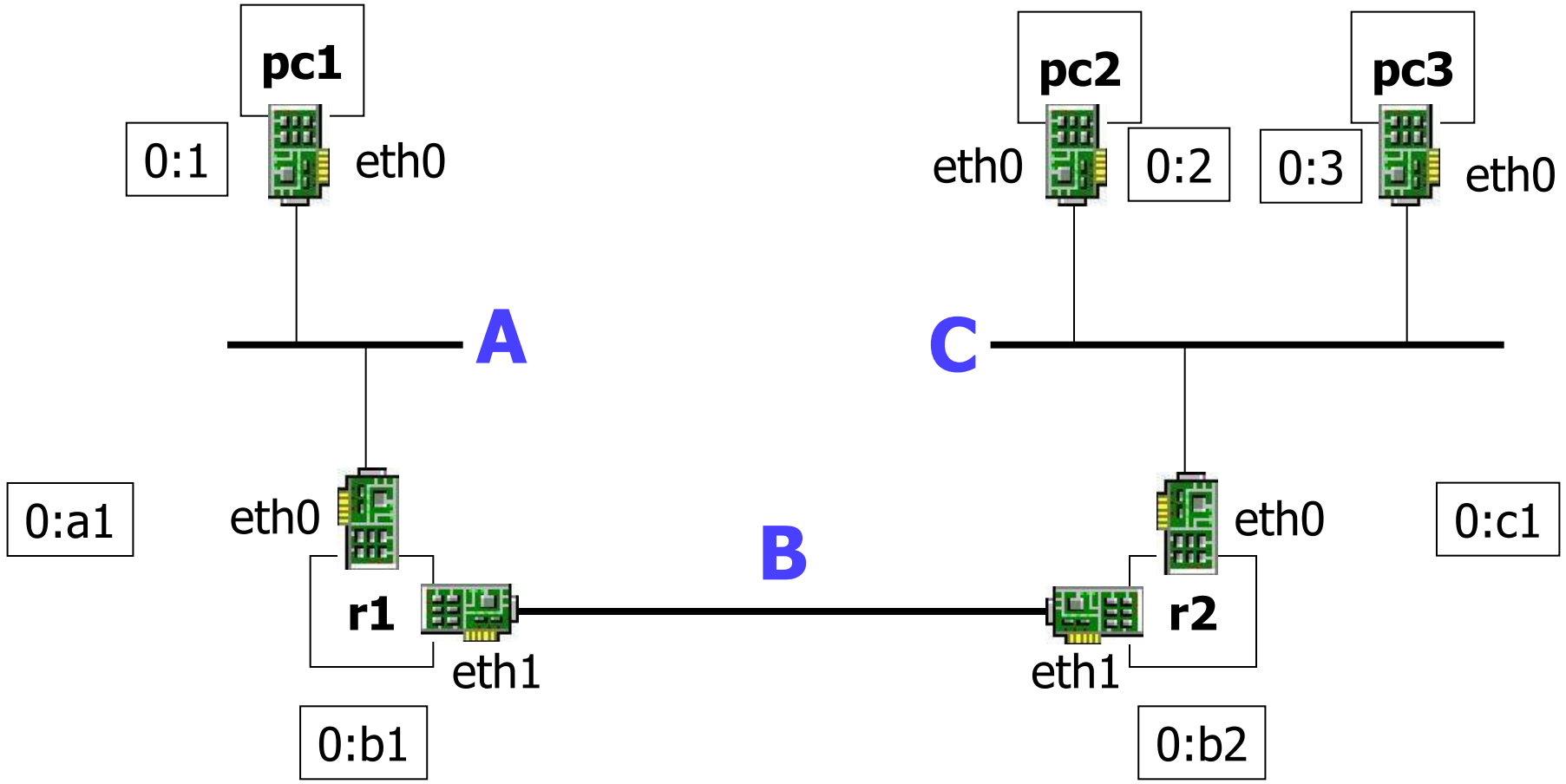


network topology – high level view



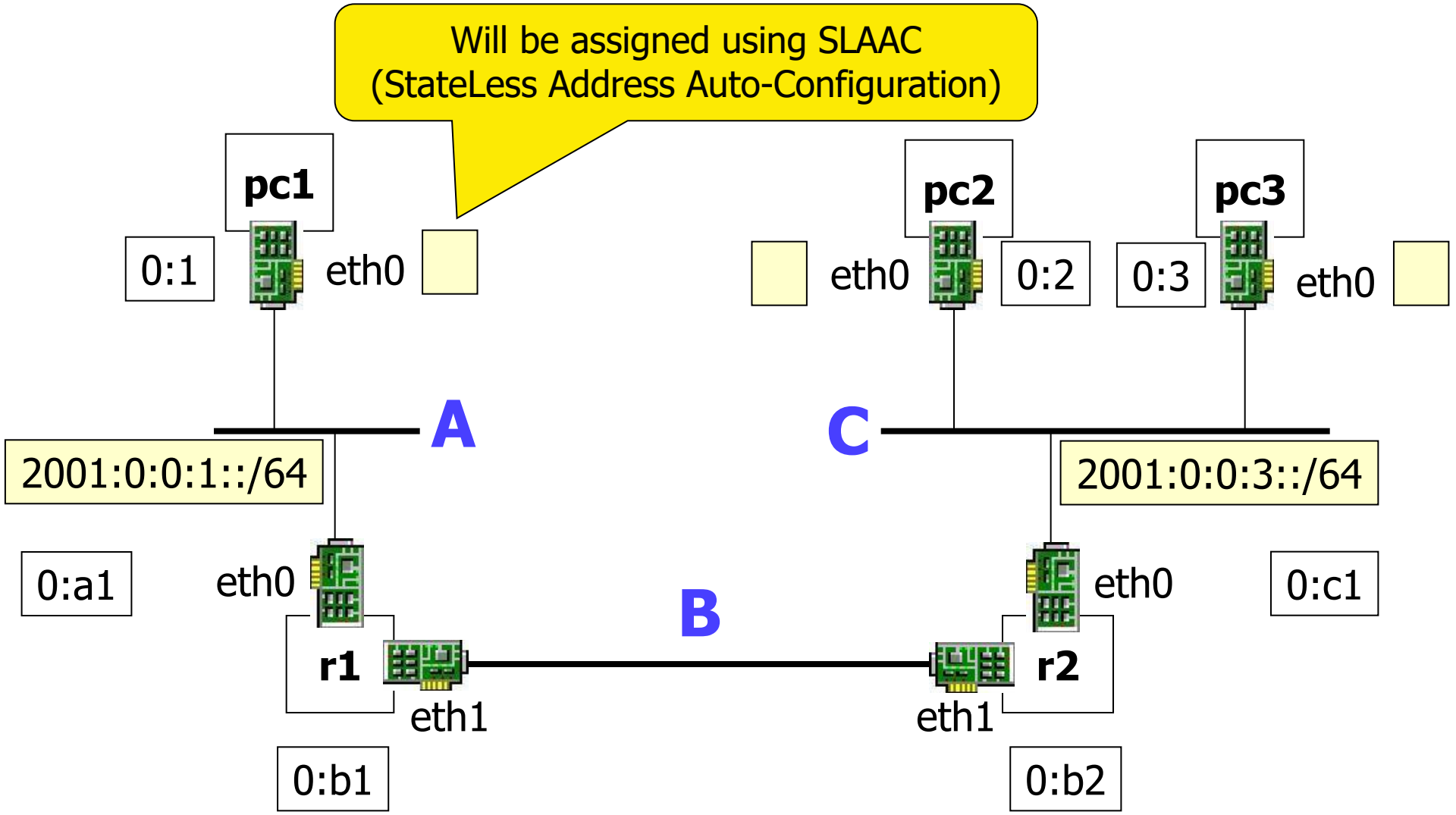
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network topology – MAC addresses

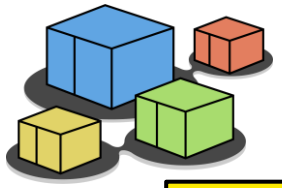


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network topology – IPv6 address plan



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a quick look at the lab

lab.conf

```
r1[0]="A/00:00:00:00:00:a1"
r1[1]="B/00:00:00:00:00:b1"
r1[image]="kathara/core"
r1[ipv6]="True"

r2[0]="C/00:00:00:00:00:c1"
r2[1]="B/00:00:00:00:00:b2"
r2[image]="kathara/core"
r2[ipv6]="True"
```

lab.conf

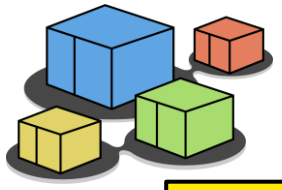
```
pc1[0]="A/00:00:00:00:00:01"
pc1[image]="kathara/core"
pc1[ipv6]="True"
pc1[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

pc2[0]="C/00:00:00:00:00:02"
pc2[image]="kathara/core"
pc2[ipv6]="True"
pc2[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

pc3[0]="C/00:00:00:00:00:03"
pc3[image]="kathara/core"
pc3[ipv6]="True"
pc3[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

wireshark[bridged]=true
wireshark[port]="3000:3000"
wireshark[image]="lscr.io/linuxserver/wireshark"
wireshark[num_terms]=0
```

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a quick look at the lab

lab.conf

```
r1[0]="A/00:00:00:00:00:a1"
r1[1]="B/00:00
r1[image
r1[ipv6]="True"

r2[0]="C/00:00:00:00:00:c1"
r2[1]="B/00:00:00:00:00:b2"
r2[image]="kathara/core"
r2[ipv6]="True"
```

Set the MAC address on the interface

lab.conf

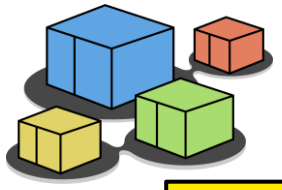
```
pc1[0]="A/00:00:00:00:00:01"
pc1[image]="kathara/core"
True"
pc1[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

pc2[0]="C/00:00:00:00:00:02"
pc2[image]="kathara/core"
pc2[ipv6]="True"
pc2[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

pc3[0]="C/00:00:00:00:00:03"
pc3[image]="kathara/core"
pc3[ipv6]="True"
pc3[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

wireshark[bridged]=true
wireshark[port]="3000:3000"
wireshark[image]="lscr.io/linuxserver/wireshark"
wireshark[num_terms]=0
```

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a quick look at the lab

lab.conf

```
r1[0]="A/00:00:00:00:00:a1"
r1[1]="B/00:00
r1[image
r1[ipv6]="True"

r2[0]="C/00:00:00:00:00:c1"
r2[1]="B/00:00:00:00:00:b2"
r2[image]="kathara/core"
r2[ipv6]="True"
```

Set the MAC address on the interface

lab.conf

```
pc1[0]="A/00:00:00:00:00:01"
pc1[image]="kathara/core"
True"
pc1[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

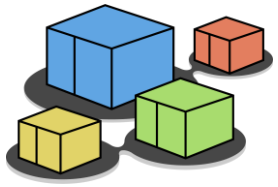
pc2[0]="C/00:00:0
pc2[image]="kathara/core"
pc2[ipv6]="True"
pc2[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

pc3[0]="C/00:00:00:00:00:03"
pc3[image]="kathara/core"
pc3[ipv6]="True"
pc3[sysctl]="net.ipv6.conf.eth0.accept_ra=2"

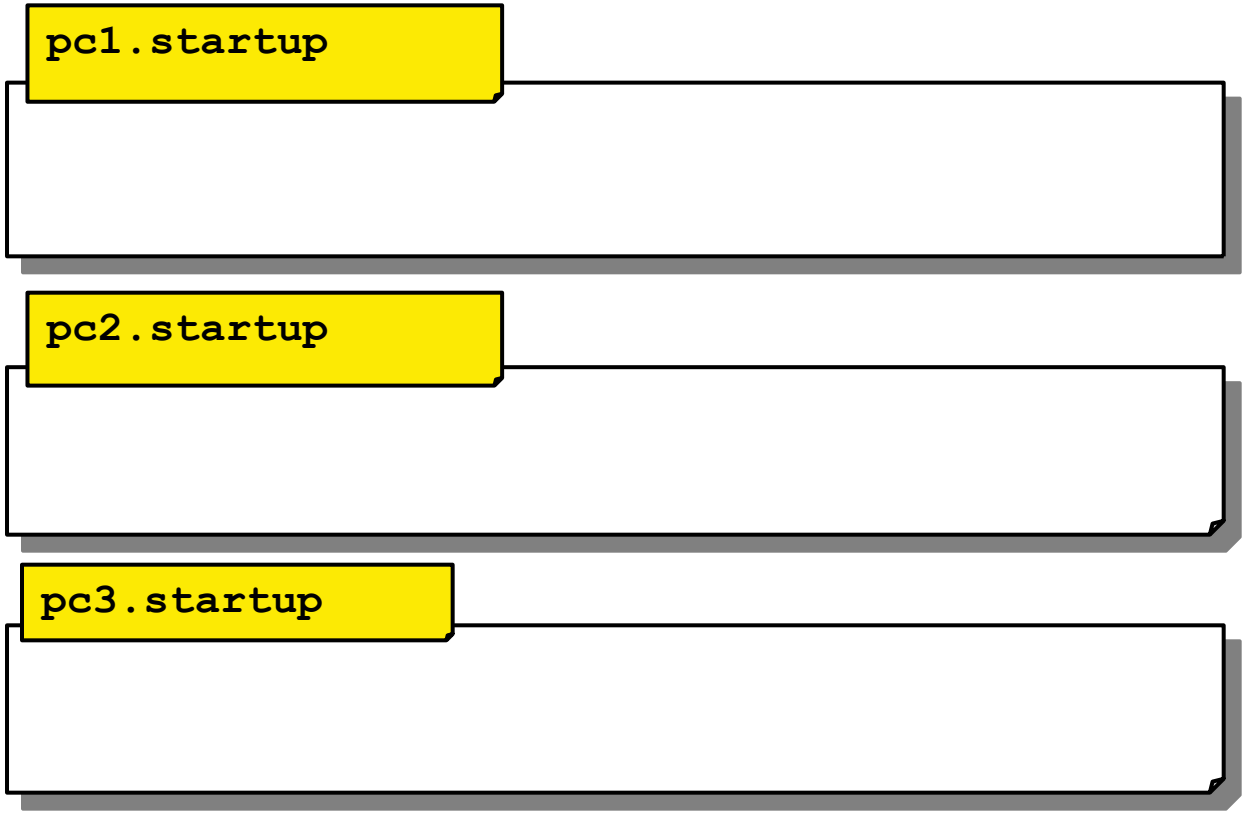
wireshark[bridged]=true
wireshark[port]="3000:3000"
wireshark[image]="lscr.io/linuxserver/wireshark"
wireshark[num_terms]=0
```

Accept router advertisements on eth0

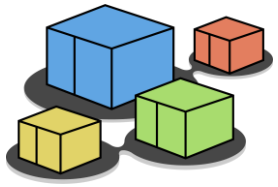
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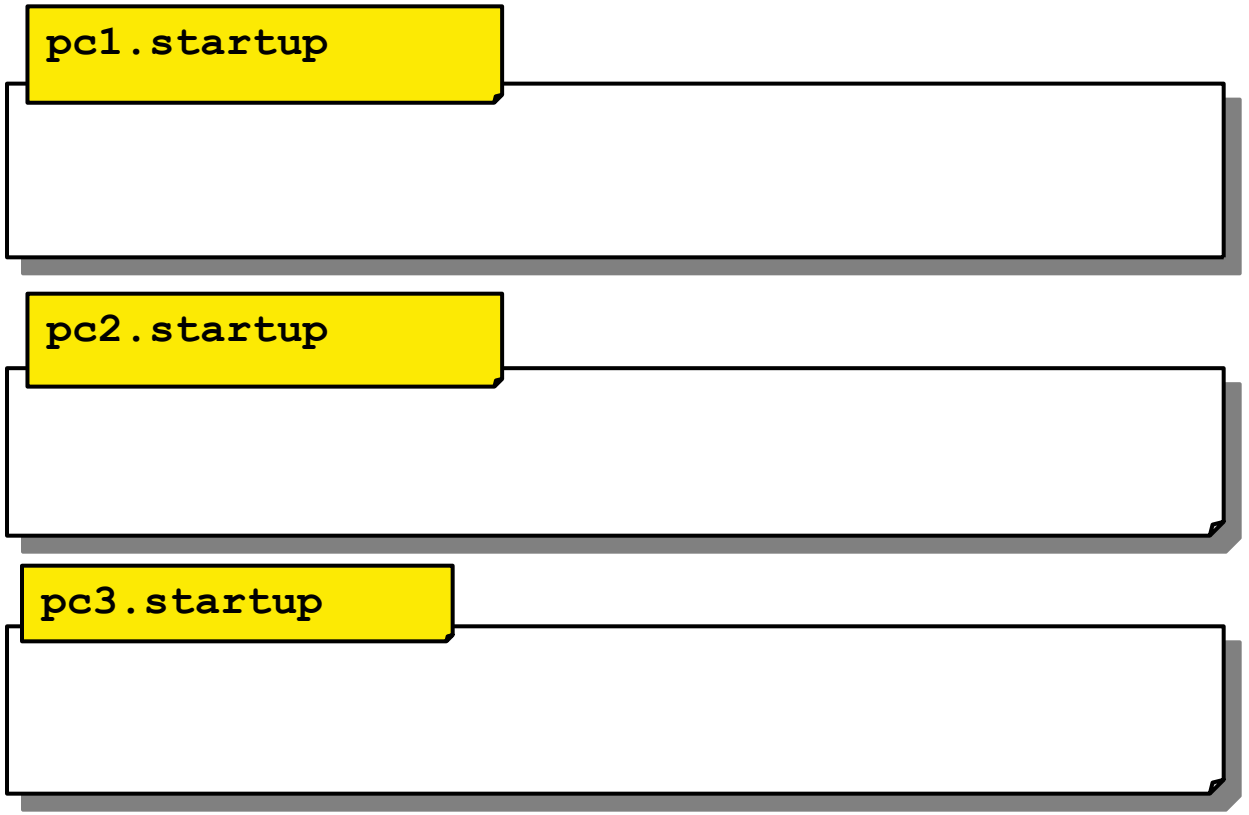


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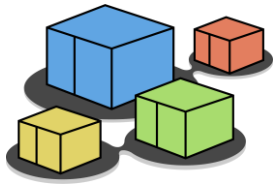


a quick look at the lab

no command is given to configure an IPv6 address or a default gateway, since they come from the **stateless auto-configuration**



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a quick look at the lab

no command is given to configure an IPv6 address or a default gateway, since they come from the **stateless auto-configuration**

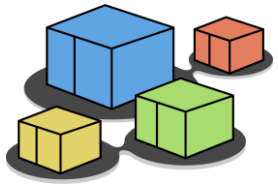
Since the .startup files for pc1, pc2, and pc3 are empty, there's no need to include them in the lab configuration

pc1.startup

pc2.startup

pc3.startup

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a quick look at the lab

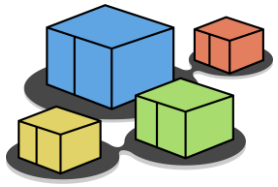
r1.startup

```
ip address add fe80::1/64 dev eth1

ip route add 2001:0:0:1::/64 dev eth0
ip route add 2001:0:0:3::/64 via fe80::2 dev eth1

chmod o-rw /etc/radvd.conf
systemctl start radvd
```

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a quick look at the lab

a static link-local address is given to eth1 to simplify routing table configuration

Otherwise, we need to know the autoconfigured link-local address of r2 (fe80::200:ff:fe00:b2)

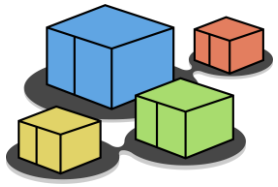
r1.startup

```
ip address add fe80::1/64 dev eth1

ip route add 2001:0:0:1::/64 dev eth0
ip route add 2001:0:0:3::/64 via fe80::2 dev eth1

chmod o-rw /etc/radvd.conf
systemctl start radvd
```

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a quick look at the lab

the routing table is set

We add an entry for each LAN. We need the `2001:0:0:1::/64` since we do not configure any global address on that LAN for r1

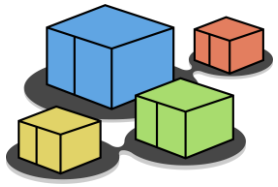
r1.startup

```
ip address add fe80::1/64 dev eth1

ip route add 2001:0:0:1::/64 dev eth0
ip route add 2001:0:0:3::/64 via fe80::2 dev eth1

chmod o-rw /etc/radvd.conf
systemctl start radvd
```

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a quick look at the lab

the correct privileges for radvd.conf are set and the radvd service is started

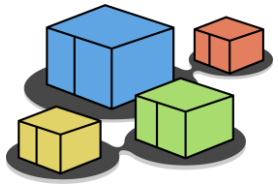
r1.startup

```
ip address add fe80::1/64 dev eth1

ip route add 2001:0:0:1::/64 dev eth0
ip route add 2001:0:0:3::/64 via fe80::2 dev eth1

chmod o-rw /etc/radvd.conf
systemctl start radvd
```

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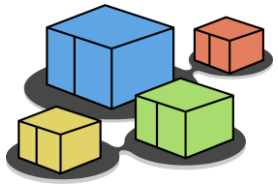
a quick look at the lab

this configuration file, of the radvd daemon, is in the /etc directory of r1

radvd.conf

```
interface eth0
{
        AdvSendAdvert on;
        MinRtrAdvInterval 3;
        MaxRtrAdvInterval 9;
        AdvDefaultLifetime 27;
        prefix 2001:0:0:1::/64 {};
};
```

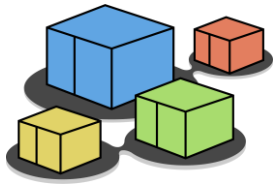
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router advertisement

- radvd is a daemon
- it is used to send router advertisement messages
- the configuration of radvd is specified in the radvd.conf file

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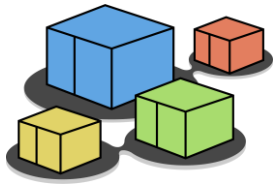
a quick look at the lab

interface of the router where the advertisements are sent

radvd.conf

```
interface eth0
{
        AdvSendAdvert on;
        MinRtrAdvInterval 3;
        MaxRtrAdvInterval 9;
        AdvDefaultLifetime 27;
        prefix 2001:0:0:1::/64 {};
};
```

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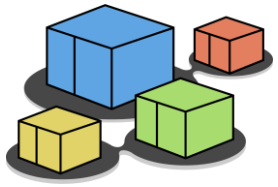
a quick look at the lab

radvd.conf

```
interface eth0
{
        AdvSendAdvert on;
        MinRtrAdvInterval 3;
        MaxRtrAdvInterval 9;
        AdvDefaultLifetime 27;
        prefix 2001:0:0:1::/64 {};
};
```

the announced prefix

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a quick look at the lab

advertisements are sent

minimum interval between consecutive advertisements

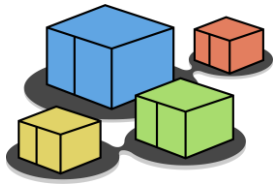
maximum interval between consecutive advertisements

time interval for default gateway validity

radvd.conf

```
interface eth0
{
        AdvSendAdvert on;
        MinRtrAdvInterval 3;
        MaxRtrAdvInterval 9;
        AdvDefaultLifetime 27;
        prefix 2001:0:0:1::/64 {};
};
```

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a quick look at the lab

r2.startup

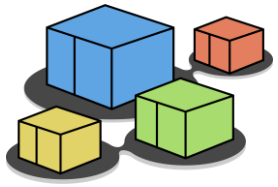
```
ip address add fe80::2/64 dev eth1

ip route add 2001:0:0:3::/64 dev eth0
ip route add 2001:0:0:1::/64 via fe80::1 dev eth1

chmod o-rw /etc/radvd.conf
systemctl start radvd
```

similar configuration for router r2

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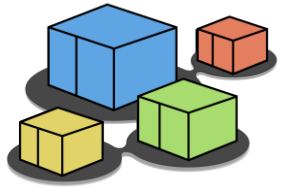


start the lab

- start the lab

```
user@localhost:~$ cd kathara-lab_basic-ipv6
user@localhost:~/kathara-lab_basic-ipv6$ lstart
```

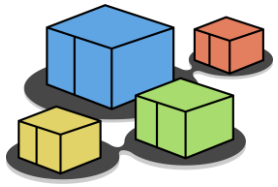
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useful commands

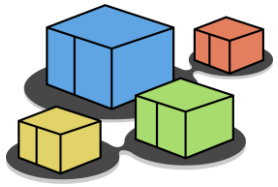
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check the IPv6 addresses

- on r1, and r2
  - perform the `ip address` command, to check the addresses assigned to the interfaces
    - the `ip -6 address` command shows only the IPv6 addresses
    - look at eth and loopback interfaces

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check the IPv6 addresses

loopback
IPv4: 127.0.0.1/8
IPv6: ::1/128

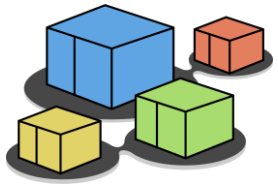
eth0
link-local: fe80::200:ff:fe00:a1/64

eth1
link-local: fe80::200:ff:fe00:b1/64
link-local: fe80::1/64

r1

```
root@r1:/# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
75: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 00:00:00:00:00:a1 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::200:ff:fe00:a1/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
79: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 00:00:00:00:00:b1 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::1/64 scope link
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe00:b1/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

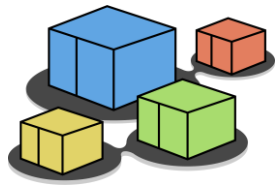
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check the router routing table

- on r1 and r2
  - perform the **`routel -6`** command, to check the routing table

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check the router routing table

Lan A

Lan C

link-local addresses on A and B

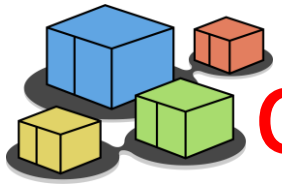
general multicast prefix

r1

```
root@r1:/# routel -6
Dst                    Gateway     Prefsrc    Protocol Scope   Dev   Table
2001:0:0:1::/64                                           eth0
2001:0:0:3::/64        fe80::2                            eth1
fe80::/64                                     kernel      eth0
fe80::/64                                     kernel      eth1
::1                                           kernel      lo    local
fe80::                                        kernel      eth1  local
fe80::                                        kernel      eth0  local
fe80::1                                       kernel      eth1  local
fe80::200:ff:fe00:a1                          kernel      eth0  local
fe80::200:ff:fe00:b1                          kernel      eth1  local
ff00::/8                                      kernel      eth0  local
ff00::/8                                      kernel      eth1  local
```

R2 link local address on B

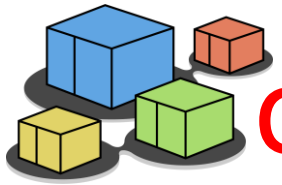
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check auto-configured IPv6 addresses

- on pc1, pc2, pc3
  - perform the `ip address` command, to check the IPv6 addresses assigned to the interfaces by the stateless auto-configuration
    - possibly, perform the `ip -6 address` command
    - look at eth and loopback interfaces

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check auto-configured IPv6 addresses

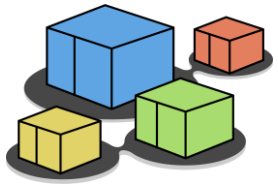
loopback
IPv4: 127.0.0.1/8
IPv6: ::1/128

eth0
global unicast: 2001::1:200:ff:fe00:1/64
link-local: fe80::200:ff:fe00:1/64

pc1

```
root@pc1:/# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
7: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
state UP group default qlen 1000
    link/ether 00:00:00:00:00:01 brd ff:ff:ff:ff:ff:ff
    inet6 2001::1:200:ff:fe00:1/64 scope global dynamic mngtmpaddr
       valid_lft 86394sec preferred_lft 14394sec
    inet6 fe80::200:ff:fe00:1/64 scope link
       valid_lft forever preferred_lft forever
```

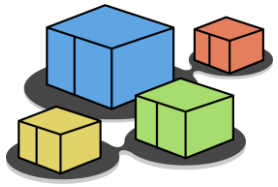
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check the default route

- on pc1, pc2, and pc3
  - perform the **`routel -6`** command, to check the presence of a default route

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check the default route

- on pc1, pc2, and pc3
  - perform the **`routel -6`** command, to check the presence of a default route

pc1

```
root@pc1:/# routel -6
Dst                    Gateway                 Prefsrc   Protocol Scope   Dev
::1                                                      kernel           lo

2001:0:0:1::                                             kernel           eth0
2001::1:200:ff:fe00:1                                    kernel           eth0
2001:0:0:1::/64                                          kernel           eth0
fe80::                                                   kernel           eth0
fe80::200:ff:fe00:1                                      kernel           eth0
fe80::/64                                                kernel           eth0
ff00::/8                                                 kernel           eth0
default                fe80::200:ff:fe00:a1              ra               eth0
```

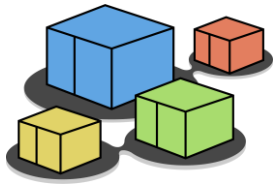
loopback prefix

default route by r1

learned by a router advertisment

link-local address of r1 on A

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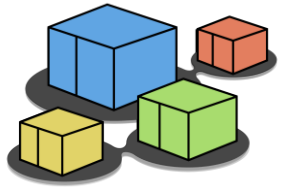
sniff the traffic

- connect the wireshark device to collision domain C

```
user@localhost:~/kathara-lab_basic-ipv6$ kathara lconfig -n wireshark --add C
```

- open any browser on the host machine
  - on **localhost:3000**
  - sniff eth1

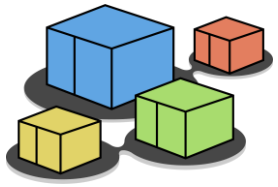
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Kathará

ping from pc3 to pc2 and related ICMPv6 behaviour

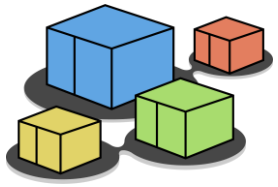
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on pc3

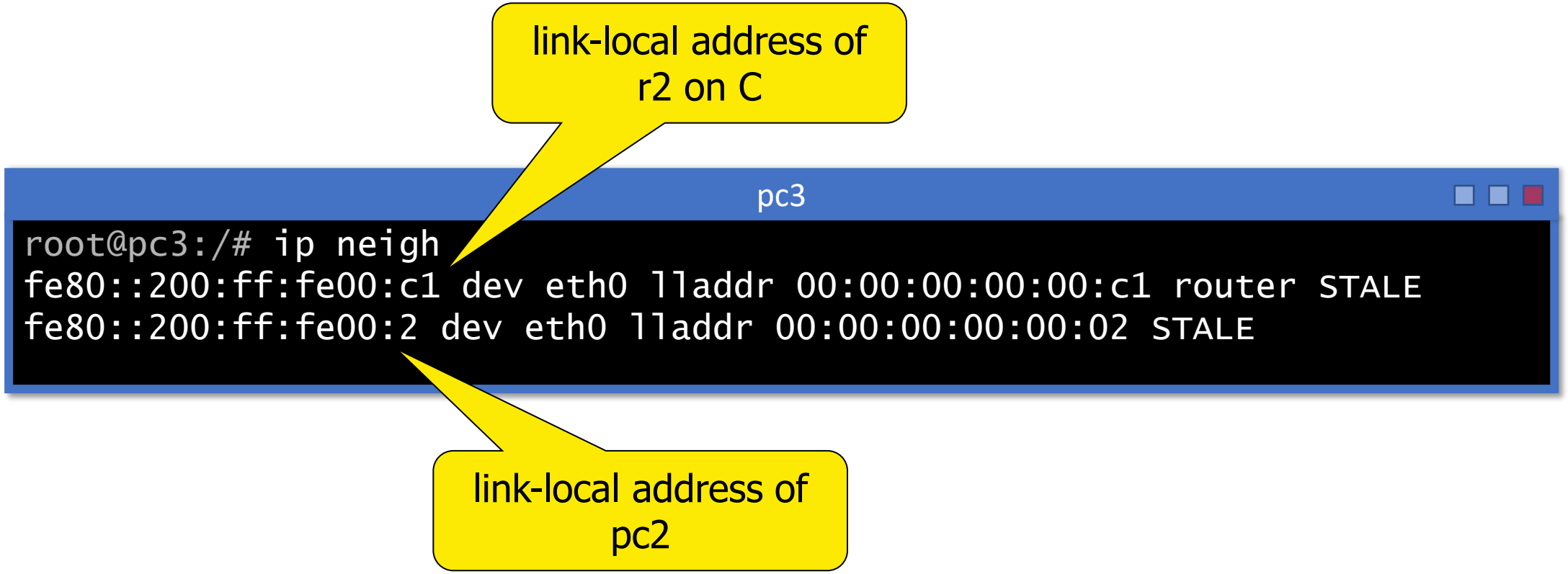
1. inspect the neighbor cache
2. execute a ping command towards pc2
3. inspect again the neighbor cache
4. give a look at the packets captured by Wireshark

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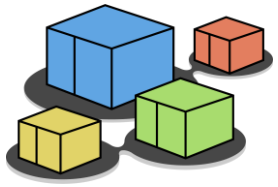


inspecting the neighbor cache

- similar to the IPv4 ARP cache



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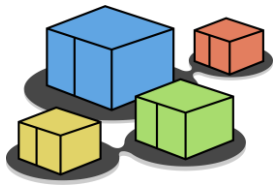
ping and inspect the neighbor cache

pc3

```
root@pc3:/# ping 2001::3:200:ff:fe00:2
PING 2001::3:200:ff:fe00:2(2001::3:200:ff:fe00:2) 56 data bytes
64 bytes from 2001::3:200:ff:fe00:2: icmp_seq=1 ttl=64 time=1.50 ms
64 bytes from 2001::3:200:ff:fe00:2: icmp_seq=2 ttl=64 time=0.585 ms
64 bytes from 2001::3:200:ff:fe00:2: icmp_seq=3 ttl=64 t
^C
--- 2001::3:200:ff:fe00:2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.585/0.953/1.500/0.394 ms
root@pc3:/# ip neigh
2001::3:200:ff:fe00:2 dev eth0 lladdr 00:00:00:00:00:02 router REACHABLE
fe80::200:ff:fe00:c1 dev eth0 lladdr 00:00:00:00:00:c1 router STALE
fe80::200:ff:fe00:2 dev eth0 lladdr 00:00:00:00:00:02 DELAY
```

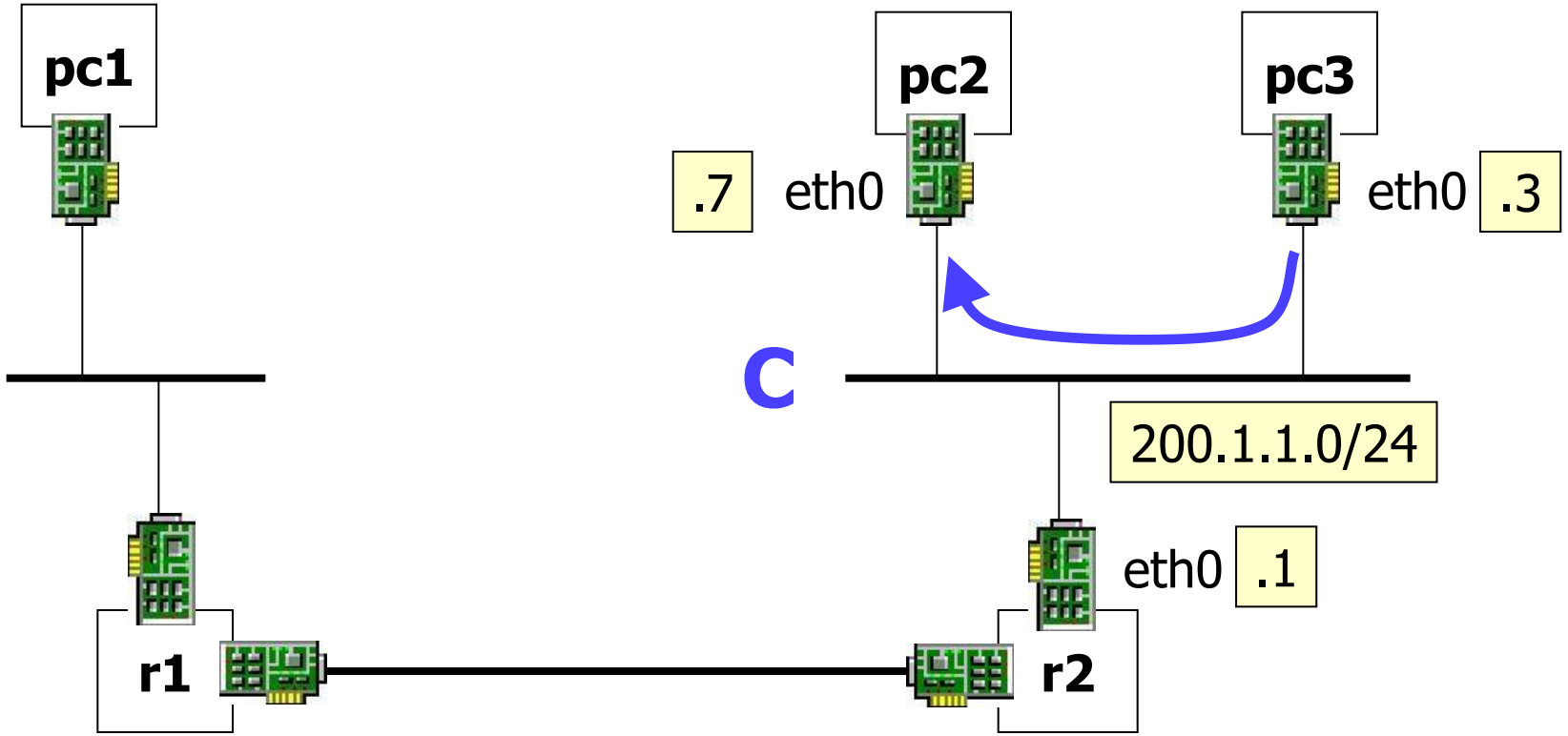
global IPv6 address of pc2

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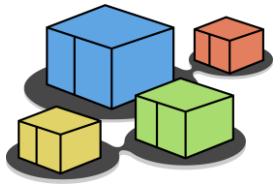


inspecting the neighbor cache

- traffic within the same network does not traverse routers



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inspecting the neighbor cache

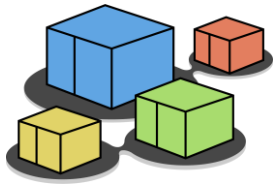
- communications are usually bi-directional
- the receiver of the neighbor solicitation learns the mac address of the other party

global IPv6 address of pc3

pc2

```
root@pc2:/# ip neigh
fe80::200:ff:fe00:3 dev eth0 lladdr 00:00:00:00:00:03 router STALE
2001::3:200:ff:fe00:3 dev eth0 lladdr 00:00:00:00:00:03 router STALE
fe80::200:ff:fe00:c1 dev eth0 lladdr 00:00:00:00:00:c1 router STALE
```

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wireshark

periodic router advertisments

*eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

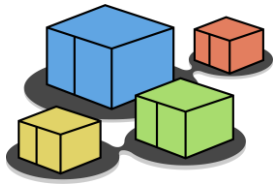
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::200:ff:fe00:c1	ff02::1	ICMPv6	110	Router Advertisement from 00:00:00:00:00:c1
2	8.851458135	fe80::200:ff:fe00:c1	ff02::1	ICMPv6	110	Router Advertisement from 00:00:00:00:00:c1
3	9.676431959	2001::3:200:ff:fe00:3	ff02::1:ff00:2	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:2 from 00:00:00:00:00:03
4	9.676594079	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Advertisement 2001::3:200:ff:fe00:2 (rtr, sol, ovr) is at 00:00:00:00:00:02
5	9.676718552	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=1, hop limit=64 (reply in 6)
6	9.676832117	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=1, hop limit=64 (request in 5)
7	10.733968276	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=2, hop limit=64 (reply in 8)
8	10.734024563	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=2, hop limit=64 (request in 7)
9	11.774587008	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=3, hop limit=64 (reply in 10)
10	11.774790187	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=3, hop limit=64 (request in 9)
11	14.734061717	fe80::200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:3 from 00:00:00:00:00:02
12	14.734322871	2001::3:200:ff:fe00:3	fe80::200:ff:fe00:2	ICMPv6	78	Neighbor Advertisement 2001::3:200:ff:fe00:3 (rtr, sol)

```
▸ Frame 1: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interfa…
▸ Ethernet II, Src: 00:00:00_00:00:c1 (00:00:00:00:00:c1), Dst: IPv6mcast_01 (33:…
▸ Internet Protocol Version 6, Src: fe80::200:ff:fe00:c1, Dst: ff02::1
▸ Internet Control Message Protocol v6
```

```
0000  33 33 00 00 00 01 00 00  00 00 00 c1 86 dd 60 00   33····· ·· ·····`·
0010  42 8e 00 38 3a ff fe 80  00 00 00 00 00 00 02 00   B··8:··· ········
0020  00 ff fe 00 00 c1 ff 02  00 00 00 00 00 00 00 00   ········ ········
0030  00 00 00 00 00 01 86 00  4a e0 40 00 00 1b 00 00   ········ J·@·····
0040  00 00 00 00 00 00 03 04  40 c0 00 01 51 80 00 00   ········ @···Q···
0050  38 40 00 00 00 00 20 01  00 00 00 00 00 03 00 00   8@···· · ········
0060  00 00 00 00 00 00 01 01  00 00 00 00 00 c1         ········ ······
```

wireshark_eth1G9ISE2.pcapng — Packets: 12 · Displayed: 12 (100.0%) — Profile: Default

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wireshark

neighbor solicitation

*eth1

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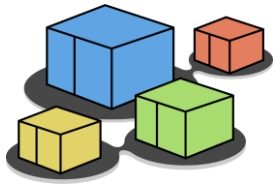
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3	9.676431959	2001::3:200:ff:fe00:3	ff02::1:ff00:2	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:2 from 00:00:00:00:00:03
4	9.676594079	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Advertisement 2001::3:200:ff:fe00:2 (rtr, sol, ovr) is at 00:00:00:00:00:02
5	9.676718552	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=1, hop limit=64 (reply in 6)
6	9.676832117	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=1, hop limit=64 (request in 5)
7	10.733968276	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=2, hop limit=64 (reply in 8)
8	10.734024563	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=2, hop limit=64 (request in 7)
9	11.774587008	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=3, hop limit=64 (reply in 10)
10	11.774790187	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=3, hop limit=64 (request in 9)
11	14.734061717	fe80::200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:3 from 00:00:00:00:00:02
12	14.734322871	2001::3:200:ff:fe00:3	fe80::200:ff:fe00:2	ICMPv6	78	Neighbor Advertisement 2001::3:200:ff:fe00:3 (rtr, sol)

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▸ Ethernet II, Src: 00:00:00_00:00:c1 (00:00:00:00:00:c1), Dst: IPv6mcast_01 (33:…
▸ Internet Protocol Version 6, Src: fe80::200:ff:fe00:c1, Dst: ff02::1
▸ Internet Control Message Protocol v6
```

```
0000  33 33 00 00 00 01 00 00  00 00 00 c1 86 dd 60 00   33····· ·· ·····`·
0010  42 8e 00 38 3a ff fe 80  00 00 00 00 00 00 02 00   B··8:··· ········
0020  00 ff fe 00 00 c1 ff 02  00 00 00 00 00 00 00 00   ········ ········
0030  00 00 00 00 00 01 86 00  4a e0 40 00 00 1b 00 00   ········ J·@·····
0040  00 00 00 00 00 00 03 04  40 c0 00 01 51 80 00 00   ········ @···Q···
0050  38 40 00 00 00 00 20 01  00 00 00 00 00 03 00 00   8@···· · ········
0060  00 00 00 00 00 00 01 01  00 00 00 00 00 c1         ········ ······
```

wireshark_eth1G9ISE2.pcapng — Packets: 12 · Displayed: 12 (100.0%) — Profile: Default

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wireshark

neighbor solicitation

neighbor advertisment

*eth1

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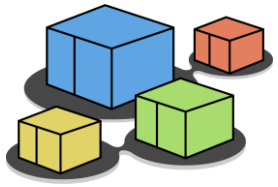
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4	9.676594079	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Advertisement 2001::3:200:ff:fe00:2 (rtr, sol, ovr) is at 00:00:00:00:00:02
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6	9.676832117	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=1, hop limit=64 (request in 5)
7	10.733968276	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=2, hop limit=64 (reply in 8)
8	10.734024563	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=2, hop limit=64 (request in 7)
9	11.774587008	2001::3:200:ff:fe00:3	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) request id=0x0004, seq=3, hop limit=64 (reply in 10)
10	11.774790187	2001::3:200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	118	Echo (ping) reply id=0x0004, seq=3, hop limit=64 (request in 9)
11	14.734061717	fe80::200:ff:fe00:2	2001::3:200:ff:fe00:3	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:3 from 00:00:00:00:00:02
12	14.734322871	2001::3:200:ff:fe00:3	fe80::200:ff:fe00:2	ICMPv6	78	Neighbor Advertisement 2001::3:200:ff:fe00:3 (rtr, sol)

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▸ Internet Control Message Protocol v6
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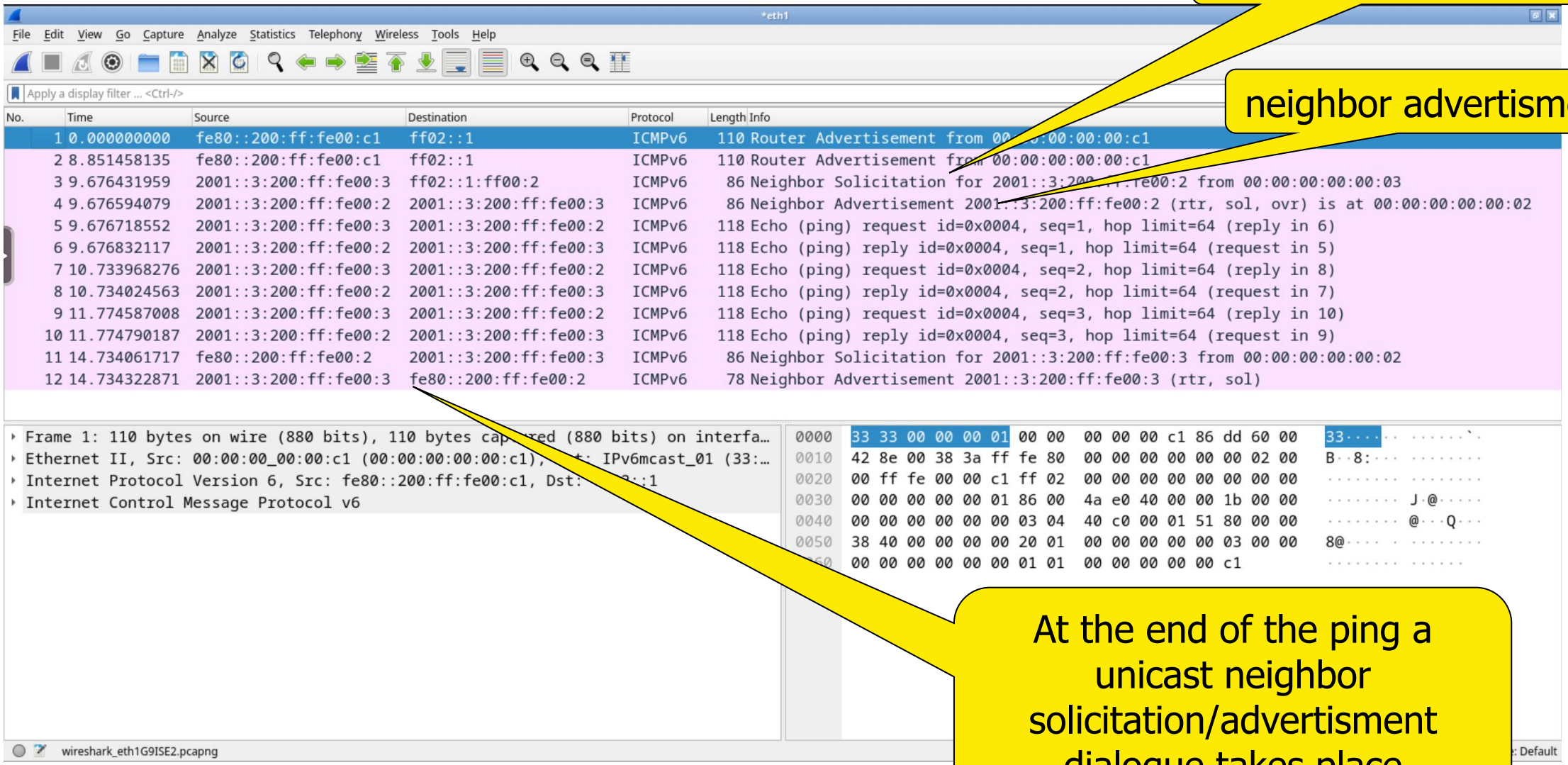
```
0000  33 33 00 00 00 01 00 00  00 00 00 c1 86 dd 60 00   33····· ·· ·····`·
0010  42 8e 00 38 3a ff fe 80  00 00 00 00 00 00 02 00   B··8:··· ········
0020  00 ff fe 00 00 c1 ff 02  00 00 00 00 00 00 00 00   ········ ········
0030  00 00 00 00 00 01 86 00  4a e0 40 00 00 1b 00 00   ········ J·@·····
0040  00 00 00 00 00 00 03 04  40 c0 00 01 51 80 00 00   ········ @···Q···
0050  38 40 00 00 00 00 20 01  00 00 00 00 00 03 00 00   8@···· · ········
0060  00 00 00 00 00 00 01 01  00 00 00 00 00 c1         ········ ······
```

wireshark_eth1G9ISE2.pcapng Packets: 12 · Displayed: 12 (100.0%) Profile: Default

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wireshark

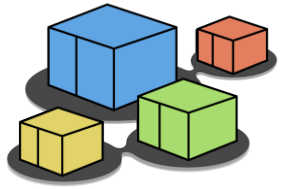


neighbor solicitation

neighbor advertisment

At the end of the ping a unicast neighbor solicitation/advertisment dialogue takes place

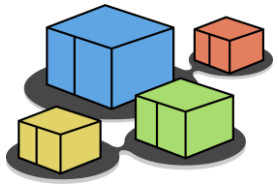
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Kathará

ping from pc2 to pc1 and related ICMPv6 behavior

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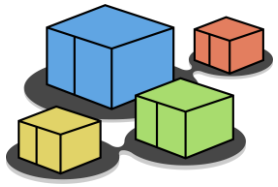
sniff the traffic

- connect the wireshark host to collision domain B

```
user@localhost:~/kathara-lab_basic-ipv6$ kathara lconfig -n wireshark --add B
```

- open any browser on the host machine
  - on **localhost:3000**
  - sniff eth2

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on pc2

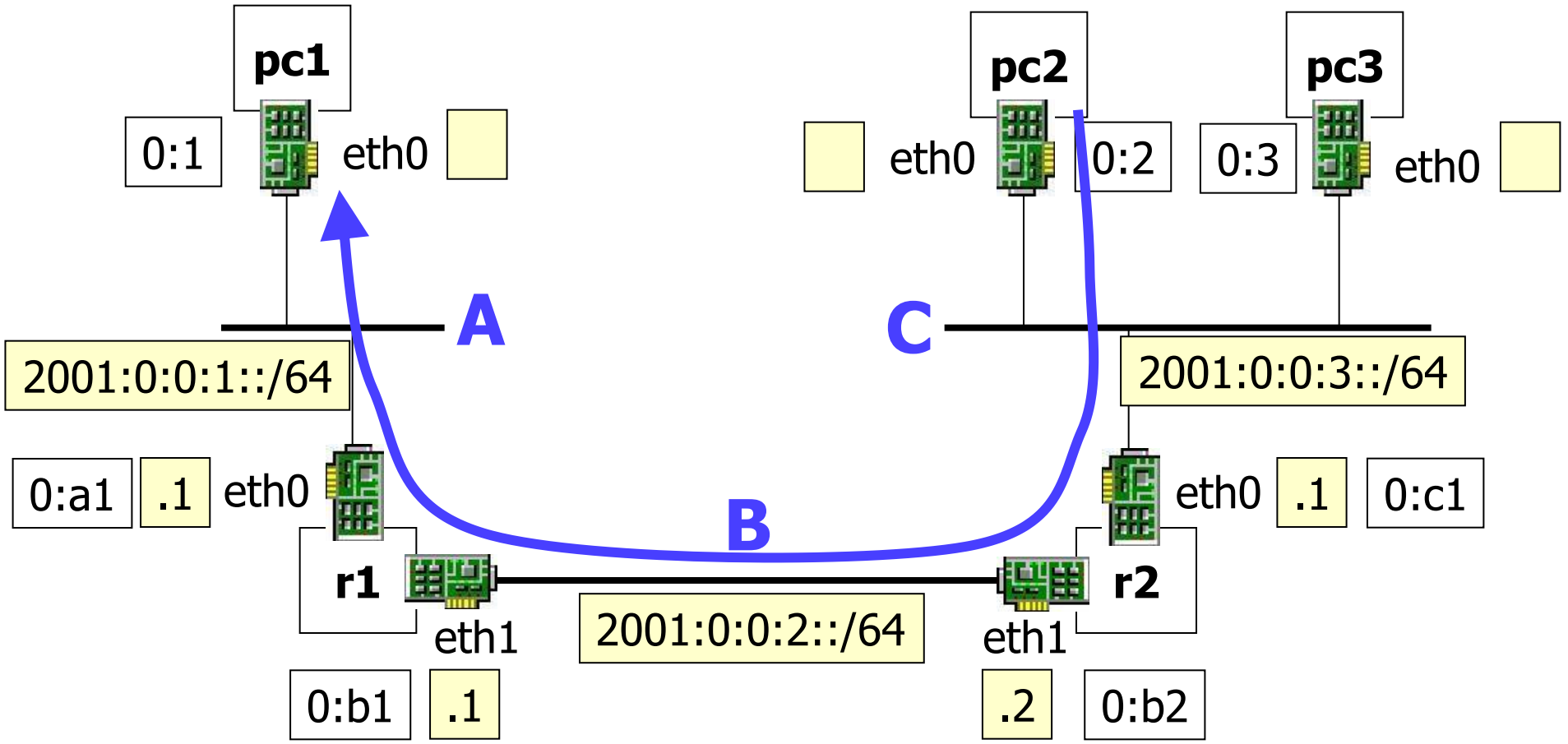
- execute a ping command towards pc1

pc2

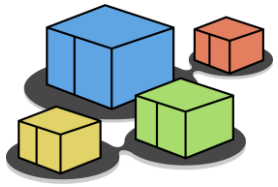
```
root@pc2:/# ping 2001::1:200:ff:fe00:1
PING 2001::1:200:ff:fe00:1(2001::1:200:ff:fe00:1) 56 data bytes
64 bytes from 2001::1:200:ff:fe00:1: icmp_seq=1 ttl=62 time=2.58 ms
64 bytes from 2001::1:200:ff:fe00:1: icmp_seq=2 ttl=62 time=1.52 ms
--- 2001::1:200:ff:fe00:1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 1.267/1.880/2.575/0.515 ms
```

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inspecting the neighbor cache



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inspecting the neighbor cache

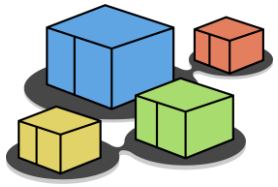
- when IP traffic is addressed outside the local network, the sender needs the mac address of the router
- ICMPv6 ND requests can get replies only within the local network

pc2

```
root@pc2:/# ip neigh
fe80::200:ff:fe00:c1 dev eth0 lladdr 00:00:00:00:00:c1 router STALE
fe80::200:ff:fe00:3 dev eth0 lladdr 00:00:00:00:00:03 STALE
```

mac address of eth0 on r2

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inspecting the neighbor cache

- what about routers?
- routers perform ND too (hence have neighbor caches)

r2 (eth1)

r1

```
root@r1:/# ip neigh
2001::1:200:ff:fe00:1 dev eth0 lladdr 00:00:00:00:00:01 router REACHABLE
fe80::200:ff:fe00:b2 dev eth1 lladdr 00:00:00:00:00:b2 router DELAY
fe80::2 dev eth1 lladdr 00:00:00:00:00:b2 router REACHABLE
fe80::200:ff:fe00:1 dev eth0 lladdr 00:00:00:00:00:01 router DELAY
fe80::200:ff:fe00:a1 dev eth0 lladdr 00:00:00:00:00:a1 router STALE
```

pc1

pc1

r1 (eth1)

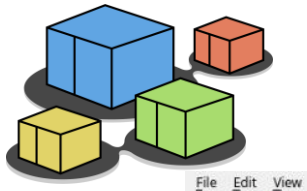
r2

```
root@r2:/# ip neigh
fe80::200:ff:fe00:c1 dev eth0 lladdr 00:00:00:00:00:c1 router STALE
fe80::200:ff:fe00:3 dev eth0 lladdr 00:00:00:00:00:03 router STALE
fe80::200:ff:fe00:2 dev eth0 lladdr 00:00:00:00:00:02 router STALE
2001::3:200:ff:fe00:2 dev eth0 lladdr 00:00:00:00:00:02 router STALE
fe80::200:ff:fe00:b1 dev eth1 lladdr 00:00:00:00:00:b1 router STALE
fe80::1 dev eth1 lladdr 00:00:00:00:00:b1 router STALE
```

pc2

pc2

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wireshark

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000…	fe80::200:ff:fe00:b2	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for fe80::1 from 00:00
2	0.00020…	fe80::1	fe80::200:ff:fe00:b2	ICMPv6	86	Neighbor Advertisement fe80::1 (rtr, sol, ov
3	0.00038…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	ICMPv6	118	Echo (ping) request id=0x0011, seq=1, hop li
4	0.00107…	fe80::200:ff:fe00:b1	ff02::1:ff00:2	ICMPv6	86	Neighbor Solicitation for fe80::2 from 00:00
5	0.00125…	fe80::2	fe80::200:ff:fe00:b1	ICMPv6	86	Neighbor Advertisement fe80::2 (rtr, sol, ov
6	0.00143…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) reply id=0x0011, seq=1, hop limi
7	1.00115…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	ICMPv6	118	Echo (ping) request id=0x0011, seq=2, hop limit=63
8	1.00173…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) reply id=0x0011, seq=2, hop limit=63 (r
9	2.00248…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	ICMPv6	118	Echo (ping) request id=0x0011, seq=3, hop limit=63
10	2.00316…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) reply id=0x0011, seq=3, hop limit=63 (in 9)
11	3.00379…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	ICMPv6	118	Echo (ping) request id=0x0011, seq=4, hop limit=63 in 12)
12	3.00439…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) reply id=0x0011, seq=4, hop limit=63 (in 11)
13	4.00504…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	ICMPv6	118	Echo (ping) request id=0x0011, seq=5, hop limit=6 in 14)
14	4.00572…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	118	Echo (ping) reply id=0x0011, seq=5, hop limit=63 in 13)
15	5.08721…	fe80::200:ff:fe00:b1	fe80::200:ff:fe00:b2	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:b2 f 00:00:00:00:00:b1
16	5.08738…	fe80::200:ff:fe00:b2	fe80::200:ff:fe00:b1	ICMPv6	78	Neighbor Advertisement fe80::200:ff:fe00:b2 (rtr sol)
17	5.09113…	fe80::200:ff:fe00:b2	fe80::200:ff:fe00:b1	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:b1 from 00:00:00:00:00:b2
18	5.09131…	fe80::200:ff:fe00:b1	fe80::200:ff:fe00:b2	ICMPv6	78	Neighbor Advertisement fe80::200:ff:fe00:b1 (rtr, sol)

which is the mac address of r1

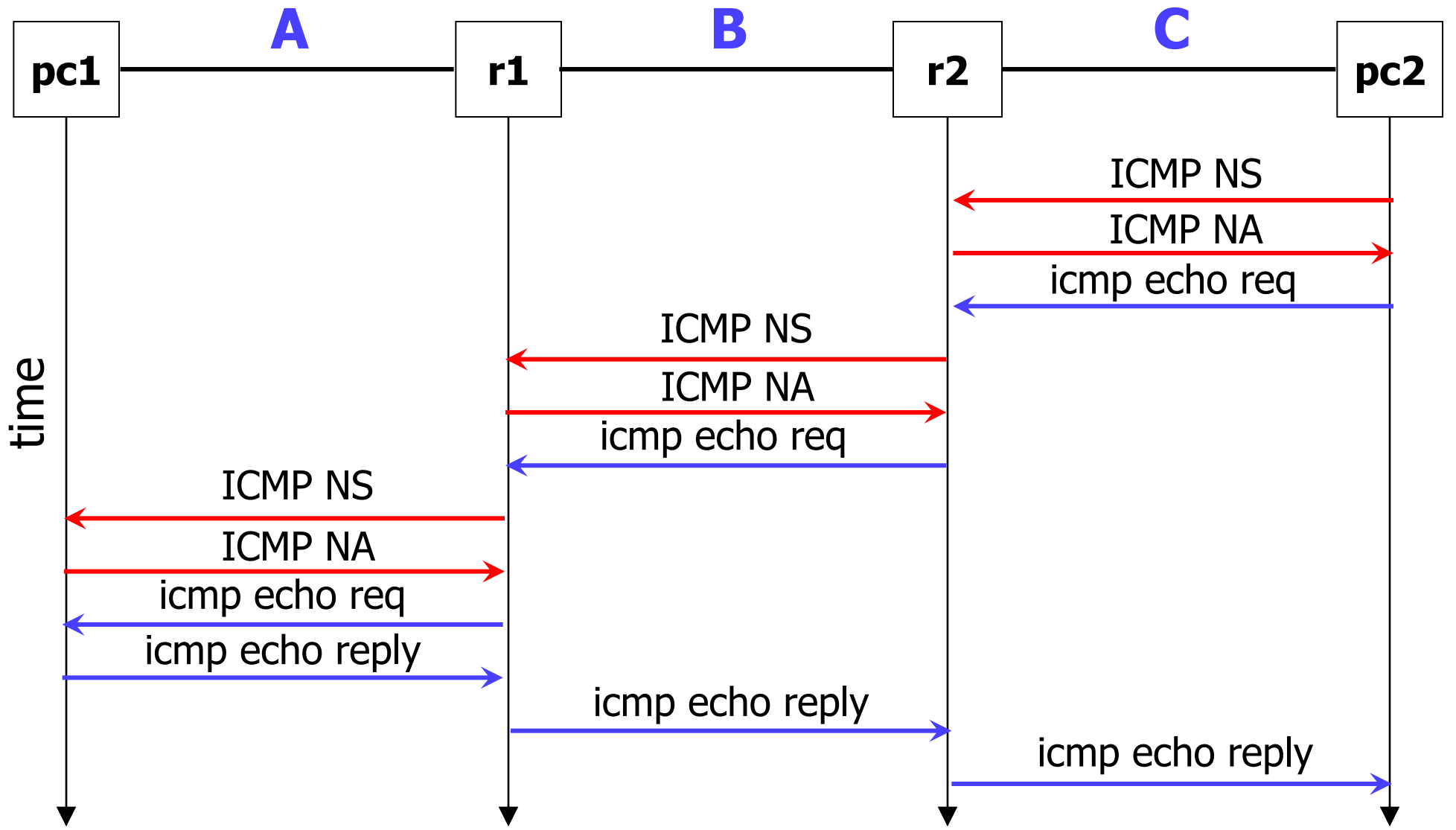
```
▸ Frame 17: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on int
▸ Ethernet II, Src: 00:00:00_00:00:b2 (00:00:00:00:00:b2), Dst: 00:00:00_00:
▸ Internet Protocol Version 6, Src: fe80::200:ff:fe00:b2, Dst: fe80::200:ff:
▾ Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0x765b [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: fe80::200:ff:fe00:b1
  ▸ ICMPv6 Option (Source link-layer address : 00:00:00:00:00:b2)
```

```
0000  00 00 00 00 00 b1 00 00  00 00 00 b2 86 dd 60 00   ······ ·· ·····`·
0010  00 00 00 20 3a ff fe 80  00 00 00 00 00 00 02 00   ··· :··· ········
0020  00 ff fe 00 00 b2 fe 80  00 00 00 00 00 00 02 00   ········ ········
0030  00 ff fe 00 00 b1 87 00  76 5b 00 00 00 00 fe 80   ······ ·· v[·····
0040  00 00 00 00 00 00 02 00  00 ff fe 00 00 b1 01 01   ······ ·· ·····
0050  00 00 00 00 00 b2                                 ·····
```

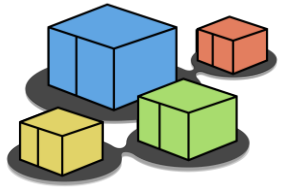
Internet Protocol Version 6 (ipv6), 40 byte(s) | Packets: 18 | Profile: Default

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understanding the whole picture



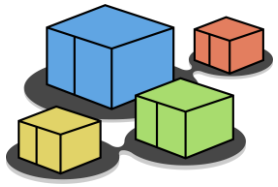
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Kathará

traceroute from pc2 to pc1 and related ICMPv6 behaviour

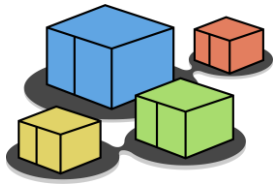
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sniff the traffic

- the wireshark host is already connected to collision domain C
- open any browser on the host machine
  - on **localhost:3000**
  - sniff eth1

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on pc2

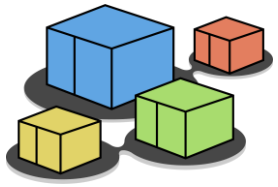
- execute a traceroute command towards pc1

Minimal time (sec. if ≤10, ms if >10) interval between probes (default 0)

pc2

```
root@pc2:/# traceroute 2001::1:200:ff:fe00:1 -z 1
traceroute to 2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1), 30 hops max, 80 byte
packets
 1  fe80::200:ff:fe00:c1 (fe80::200:ff:fe00:c1)  0.517 ms  0.471 ms  0.416 ms
 2  * * *
 3  2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1)  1.112 ms  1.087 ms  1.124 ms
```

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on pc2

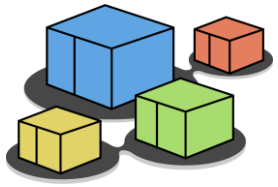
- execute a traceroute command towards pc1

eth0 of r2

pc2

```
root@pc2:/# traceroute 2001::1:200:ff:fe00:1 -z 1
traceroute to 2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1), 30 hops max, 80 byte
packets
 1  fe80::200:ff:fe00:c1 (fe80::200:ff:fe00:c1)  0.517 ms  0.471 ms  0.416 ms
 2  * * *
 3  2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1)  1.112 ms  1.087 ms  1.124 ms
```

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on pc2

- execute a traceroute command towards pc1

eth0 of r2

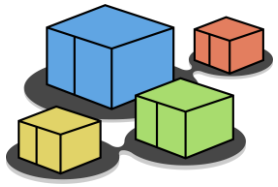
No answer from r1

Why?

pc2

```
root@pc2:/# traceroute 2001::1:200:ff:fe00:1 -z 1
traceroute to 2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1), 30 hops max, 80 byte
packets
 1  fe80::200:ff:fe00:c1 (fe80::200:ff:fe00:c1)  0.517 ms  0.471 ms  0.416 ms
 2  * * *
 3  2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1)  1.112 ms  1.087 ms  1.124 ms
```

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on pc2

- execute a traceroute command towards pc1

eth0 of r2

No answer from r1

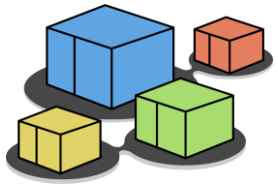
Why?

r1 does not have any global address to communicate outside its LANs

pc2

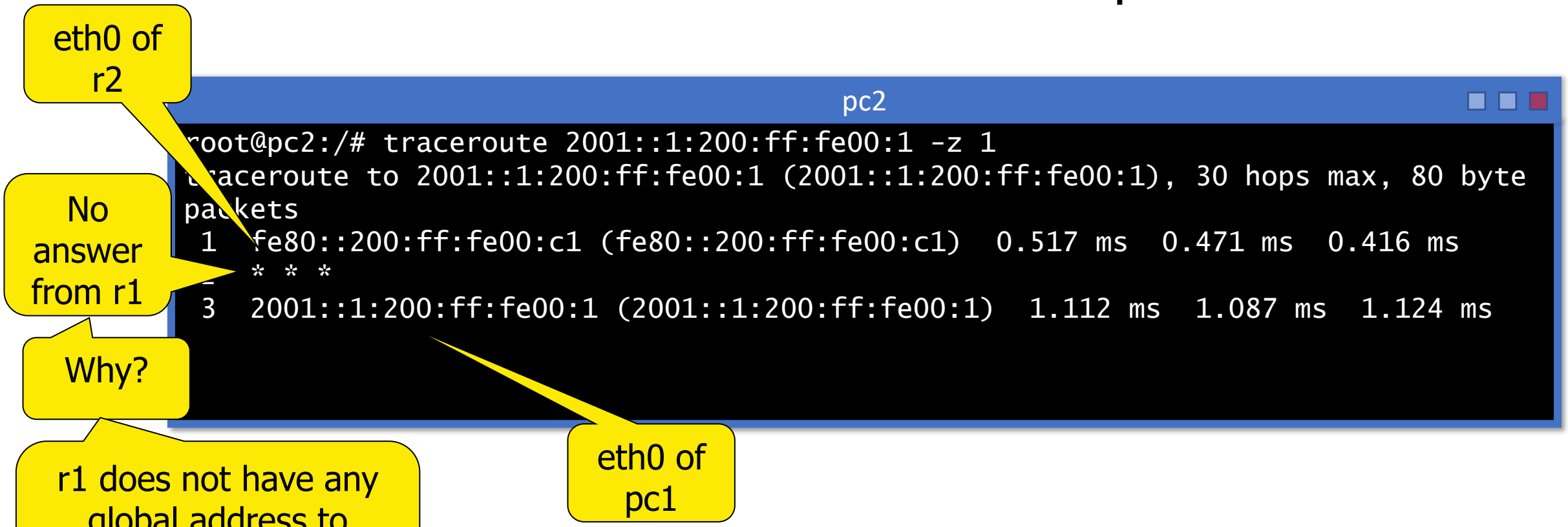
```
root@pc2:/# traceroute 2001::1:200:ff:fe00:1 -z 1
traceroute to 2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1), 30 hops max, 80 byte
packets
 1  fe80::200:ff:fe00:c1 (fe80::200:ff:fe00:c1)  0.517 ms  0.471 ms  0.416 ms
 2  * * *
 3  2001::1:200:ff:fe00:1 (2001::1:200:ff:fe00:1)  1.112 ms  1.087 ms  1.124 ms
```

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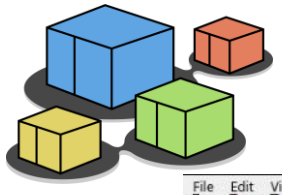


on pc2

- execute a traceroute command towards pc1



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wireshark

udp packet and corresponding ICMP Time-to-live exceeded

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No.	Time	Source	Destination	Protocol	Length	Info
1	0.0000…	fe80::200:ff:fe00:c1	ff02::1	ICMPv6	110	Router Advertisement from 00:00:00:00:00:c1
2	2.0013…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	49595 → 33434 Len=32
3	2.0015…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv6	142	Time Exceeded (Hop limit exceeded in transit)
4	3.0013…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	52894 → 33435 Len=32
5	3.0015…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv6	142	Time Exceeded (Hop limit exceeded in transit)
6	4.0014…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	39629 → 33436 Len=32
7	4.0016…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv6	142	Time Exceeded (Hop limit exceeded in transit)
8	5.0015…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	34644 → 33437 Len=32
9	6.0026…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	46229 → 33438 Len=32
10	6.3427…	fe80::200:ff:fe00:c1	ff02::1	ICMPv6	110	Router Advertisement from 00:00:00:00:00:c1
11	7.0035…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	48451 → 33439 Len=32
12	7.0113…	fe80::200:ff:fe00:2	fe80::200:ff:fe00:c1	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:c1 from 00:00:00:00:00:02
13	7.0115…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:2 from 00:00:00:00:00:c1
14	7.0117…	fe80::200:ff:fe00:c1	fe80::200:ff:fe00:2	ICMPv6	78	Neighbor Advertisement fe80::200:ff:fe00:c1 (rtr, sol)
15	7.0119…	2001::3:200:ff:fe00:2	fe80::200:ff:fe00:c1	ICMPv6	78	Neighbor Advertisement 2001::3:200:ff:fe00:2 (rtr, sol)
16	8.0043…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	48052 → 33440 Len=32
17	8.0052…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
18	9.0044…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	51519 → 33441 Len=32
19	9.0053…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
20	10.004…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	55811 → 33442 Len=32
21	10.005…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
22	12.127…	fe80::200:ff:fe00:c1	fe80::200:ff:fe00:2	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:2 from 00:00:00:00:00:c1

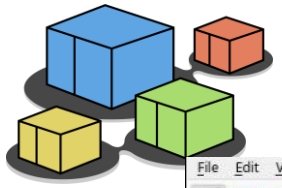
```
▸ Frame 2: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on int
▸ Ethe                00:02 (00:00:00:00:00:02), Dst: 00:00:00_00
▾ Inte                rc: 2001::3:200:ff:fe00:2, Dst: 2001::1:200
    01
  ▸ ..                . .... .... = Traffic Class: 0x00 (DSCP: CS(
    .... 11           10 1101 0000 1010 = Flow Label: 0xced0a
    Payload             gth: 40
    Next Header: UDP (17)
    Hop Limit: 1
  ▸ Source Address: 2001::3:200:ff:fe00:2
  ▸ Destination Address: 2001::1:200:ff:fe00:1
    [Source Teredo Server IPv4: 0.0.0.3]
    [Source Teredo Port: 65280]
    [Source Teredo Client IPv4: 1.255.255.253]
    [Destination Teredo Server IPv4: 0.0.0.1]
```

TTL = 1

```
0000  00 00 00 00 00 c1 00 00  00 00 00 02 86 dd 60 0c  ·····`·
0010  ed 0a 00 28 11 01 20 01  00 00 00 00 00 03 02 00  ···(·· ·
0020  00 ff fe 00 00 02 20 01  00 00 00 00 00 01 02 00  ····· ·
0030  00 ff fe 00 00 01 c1 bb  82 9a 00 28 84 3a 40 41  ·····  ···(·:@A
0040  42 43 44 45 46 47 48 49  4a 4b 4c 4d 4e 4f 50 51  BCDEFGHI JKLMNOPQ
0050  52 53 54 55 56 57 58 59  5a 5b 5c 5d 5e 5f        RSTUVWXY Z[\]^_
```

Frame (frame), 94 byte(s) Packets: 47 Profile: Default

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wireshark

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination			
1	0.0000…	fe80::200:ff:fe00:c1	ff02::1			0:00:00:c1
2	2.0013…	2001::3:200:ff:fe00:2	2001::1:200:ff			
3	2.0015…	fe80::200:ff:fe00:c1	2001::3:200:ff			in transit)
4	3.0013…	2001::3:200:ff:fe00:2	2001::1:200:ff			
5	3.0015…	fe80::200:ff:fe00:c1	2001::3:200:ff			d in transit)
6	4.0014…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	39629 → 33436 Len=32
7	4.0016…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv	142	Time Exceeded (Hop limit exceeded in transit)
8	5.0015…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	34644 → 33437 Len=32
9	6.0026…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	46229 → 33438 Len=32
10	6.3427…	fe80::200:ff:fe00:c1	ff02::1	ICMPv6	110	Router Advertisement from 00:00:00:00:00:c1
11	7.0035…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	48451 → 33439 Len=32
12	7.0113…	fe80::200:ff:fe00:2	fe80::200:ff:fe00:c1	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:c1 from 00:00:00:00:00:02
13	7.0115…	fe80::200:ff:fe00:c1	2001::3:200:ff:fe00:2	ICMPv6	86	Neighbor Solicitation for 2001::3:200:ff:fe00:2 from 00:00:00:00:00:c1
14	7.0117…	fe80::200:ff:fe00:c1	fe80::200:ff:fe00:2	ICMPv6	78	Neighbor Advertisement fe80::200:ff:fe00:c1 (rtr, sol)
15	7.0119…	2001::3:200:ff:fe00:2	fe80::200:ff:fe00:c1	ICMPv6	78	Neighbor Advertisement 2001::3:200:ff:fe00:2 (rtr, sol)
16	8.0043…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	48052 → 33440 Len=32
17	8.0052…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
18	9.0044…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	51519 → 33441 Len=32
19	9.0053…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
20	10.004…	2001::3:200:ff:fe00:2	2001::1:200:ff:fe00:1	UDP	94	55811 → 33442 Len=32
21	10.005…	2001::1:200:ff:fe00:1	2001::3:200:ff:fe00:2	ICMPv6	142	Destination Unreachable (Port unreachable)
22	12.127…	fe80::200:ff:fe00:c1	fe80::200:ff:fe00:2	ICMPv6	86	Neighbor Solicitation for fe80::200:ff:fe00:2 from 00:00:00:00:00:c1

Where are the corresponfing Time Exceed packets?

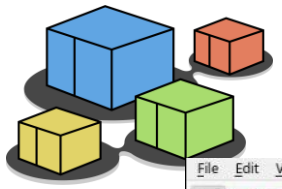
```
▸ Frame 8: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on int
▸ Ethernet II, Src: 00:00:00_00:00:02 (00:00:00:00:00:02), Dst: 00:00:00_00
▾ Int                                Src: 2001::3:200:ff:fe00:2, Dst: 2001::1:200
    01
  ▸ ..                                .... .... = Traffic Class: 0x00 (DSCP: CS(
    .... .                                    11 = Flow Label: 0x754b3
    Payload                                gth: 40
    Next Hea                               er: UDP (17)
    Hop Limit: 2
  ▸ Source Address: 2001::3:200:ff:fe00:2
  ▸ Destination Address: 2001::1:200:ff:fe00:1
    [Source Teredo Server IPv4: 0.0.0.3]
    [Source Teredo Port: 65280]
    [Source Teredo Client IPv4: 1.255.255.253]
    [Destination Teredo Server IPv4: 0.0.0.1]
```

TTL = 2

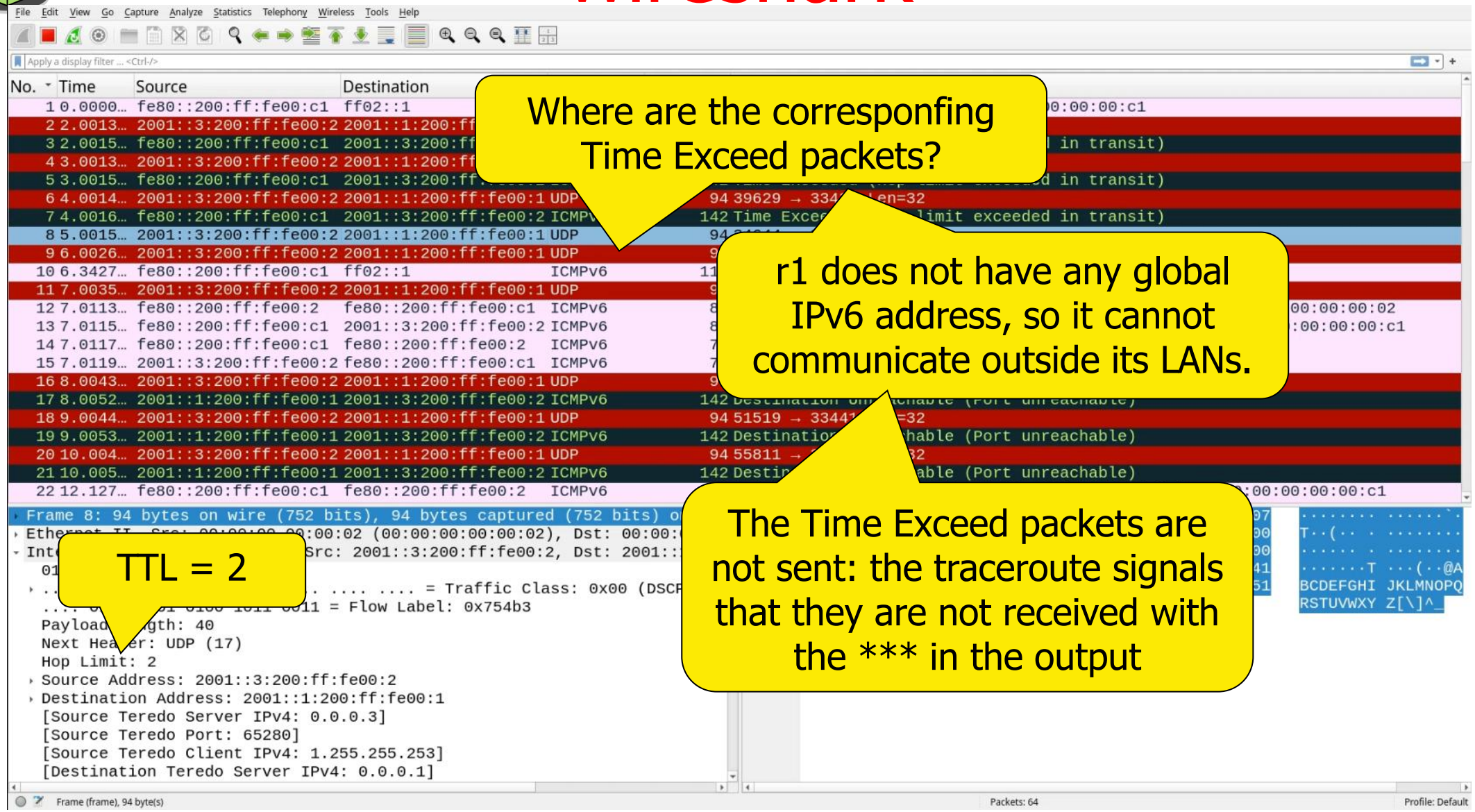
```
0000  00 00 00 00 00 c1 00 00  00 00 00 02 86 dd 60 07   ........ ......`.
0010  54 b3 00 28 11 02 20 01  00 00 00 00 00 03 02 00   T..(.. . ........
0020  00 ff fe 00 00 02 20 01  00 00 00 00 00 01 02 00   ...... . ........
0030  00 ff fe 00 00 01 87 54  82 9d 00 28 be 9e 40 41   .......T ...(..@A
0040  42 43 44 45 46 47 48 49  4a 4b 4c 4d 4e 4f 50 51   BCDEFGHI JKLMNOPQ
0050  52 53 54 55 56 57 58 59  5a 5b 5c 5d 5e 5f        RSTUVWXY Z[\]^_
```

Frame (frame), 94 byte(s) Packets: 64 Profile: Default

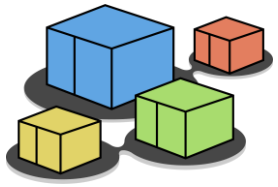
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proposed exercises

- check the different error messages obtained by trying to ping an unreachable destination in the case of
  - local destination
  - non-local destination
- which packets are exchanged in the local collision domain in the two cases?

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